



A fixed smooth spherical shell has centre O and radius a . A particle of mass m moves in complete vertical circles on the smooth inner surface of the shell, where the plane of the circular motion is vertical and passes through O . The particle has speed v when it is at point A , where OA makes an angle θ with the upward vertical through O , and $\cos \theta = \frac{1}{18}$ (see diagram).

- (a) Show that $v \geq \frac{1}{3}\sqrt{26ag}$. [5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple horizontal rows, each defined by two parallel dashed lines. The lines are evenly spaced and extend across the entire width of the page, providing a guide for letter height and placement. There is no text or other markings on the page.



It is given that $v = \frac{1}{3}\sqrt{26ag}$.

- (b)** Find, in terms of m and g , an expression for the greatest possible value of the normal reaction between the shell and the particle. [3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.